

Gordon “Reece” Appling

reeceappling@gmail.com

(704)984-2117

[linkedin.com/in/reeceappling](https://www.linkedin.com/in/reeceappling)

links.reece.appli.ng reece.appli.ng/cv

SUMMARY

Senior Software Engineer and Technical Lead with 15+ years of experience. Well-versed in designing and operating high-performance, distributed cloud systems. B.S. in Nuclear Engineering with a Minor in Mathematics. Specializes in large-scale data systems, high-throughput compute pipelines, GPU acceleration, and cost-efficient cloud architecture. Proven track record of improving reliability to six-nines availability and reducing infrastructure spend by 64% while increasing performance and stability.

WORK EXPERIENCE

Senior Software Engineer, Consultant Source Allies Inc.

May 2022 – Current

- John Deere, Images Team - Software Engineer (2022-23), Senior SWE (2023-now), Tech Lead (2025-now)

- Most senior consulting engineer on a high-performance, global-scale, team of 4-8 at John Deere’s Intelligent Solutions Group; responsible for architecture, implementation, testing, optimization, and support of a complex set of diverse cloud services utilizing geospatiotemporal agribusiness data
- Architected, implemented, and maintained cloud infrastructure and services for ingest, distributed processing, storage, manipulation, and retrieval of data for, datastores totalling over 50PB
- Created multiple ECS clusters for use in, and consuming data from, John Deere AI platforms
- Designed CUDA C/C++ Kernels used through CGo for statistics and image processing via GPU
- Improved a mission-critical image manipulation API from 65% reliability to 99.9999% success rate
- Spearheaded implementation of an ECS cluster using an advanced topology algorithm (from a PhD thesis), achieving >100x performance gains over its original implementation on TB-scale datasets
- Built a Go-based compiler that transformed nested JSON instructions into machine-executable operations across EC2 clusters for retrieving and manipulating geospatial agricultural data
- Wrote a compiler in Go which resolved nested JSON instructions to machine-executable instructions for retrieving and manipulating geospatial agricultural data within EC2 clusters
- Saved \$18M of a \$28M budget (64%) in 2024, while still increasing service stability and throughput
- Ensured maximum service uptime via careful design and rollout of CI/CD pipelines operated via self-hosted GitHub Actions runners, in conjunction with Infrastructure as Code via Terraform
- Setup monitoring, dashboards, alerting, traces, and profiling (Datadog/Grafana/CloudWatch)
- Responsible for educating engineers on infrastructure, codebase, domain, and best practices
- Utilized primarily Go, Terraform, Bash, and Docker on AWS, but also used Scala, Github Actions, C/C++ with CUDA, DroneCI, python, javascript, typescript, Kotlin, and more
- Platforms utilized: AWS (>30 separate services), Datadog, LogCentral, Rally, Azure DevOps, Github, DroneCI, Grafana, Prometheus, Confluence, and more

- Simpson University: Software Engineer, Consultant

- Upgraded the University’s payment and donation gateway. Remediated resulting bugs.

- Source Allies: Software Engineer, Consultant

- Upgraded company internal payment gateway to a newer version of Java Spring.
- Secured all company machines via JAMF to ensure the protection of company and client data.
- Designed and created a Slack bot integration with Small Improvements to automate monthly announcements, and employee creation and completion of personal and professional goals.
- Redesigned Jira workflows streamlining the hiring process and onboarding systems for remote coworkers.

Software Engineer - Freelance, Remote

Jan 2012 to May 2022

Applying Technologies

- Designed and created programs and systems to achieve a desired outcome utilizing a variety of languages and tools. Also providing long term Systems Administration and IT support as needed.

- ArrowNail LLC

- Used serverless services on AWS to support a React geospatial web app, Node API via lambda functions, and an aurora database. Provided IT and Systems Administration Support.
- Set up CI/CD pipeline in Gitlab CI to make future deployments seamless.

- CritCola

- Consulted on hosting game servers and discord bots for a large online community such that they could be deployed or destroyed, on short notice with persistent game data utilizing GitLabCI, Terraform, CloudFlare, and AWS (EC2, EBS, IAM).
- Created a final product with a spin-up time of approximately 3 minutes.

- Masters Wastewater Statistics Research

- Created programs for a student doing research for his Masters Degree in Public Health. Provided data on E.Coli samples from different waterways, the programs checked the statistical validity on different E.Coli indicating kits.

- Well Aware NC

- Designed, created, and hosted a website for Well Aware NC, a University of North Carolina Chapel Hill affiliated nonprofit focused on the testing of well water contaminants within North Carolina.

- Wildlife Mapping with R

- Utilized spatiotemporal data for wildlife in a specified area over a specified date range in order to produce population density maps.

Civil/Structural Engineer and Tower Climber - Raleigh, NC

Jun 2018 to May 2020

Talley Associates of Engineering, Inc. and TEI

- Designed and analyzed structures to check against company, customer, and governmental criteria.
- Surveyed structural materials, equipment, dimensions, and physical states to replicate and analyze.
- Certified as an authorized cell tower climber and rescuer through Comtrain USA.

EDUCATION

- **North Carolina State University**
Bachelor of Science (B.S.), Nuclear Engineering, Minor in Mathematics

CERTIFICATIONS

- AWS Certified Solutions Architect - Associate

TECHNICAL SKILLS

- **AWS:** EC2, RDS, SQS, ECR, IAM, VPC, CloudFormation, S3, EFS, CloudWatch, Route53, Lambda, CloudFront, DynamoDB/DAX, SNS, SES, ElastiCache, Kinesis, Fargate, Autoscaling, EventBridge, Athena, API Gateway, System Manager, SAM, CodeCommit, CodeDeploy, Chime, AMI
- **Languages:** Go, Scala, Node.js, Typescript, C, C++, Bash, Python, FORTRAN, Java, PHP, R, SQL and more.
- **Web Technologies:** HTML5, CSS3, React, WASM, jQuery, GraphQL
- **CI/CD Tools:** Github Actions, GitLab CI/CD, DroneCI, AWS CodeDeploy
- **Infrastructure:** Terraform, AWS CloudFormation, AWS SAM
- **Containerization:** Docker, Docker Compose, ECR, ECS, Fargate, PodMan
- **Monitoring Tools:** CloudWatch, Datadog, OTel, Grafana, LogCentral
- **Databases:** MongoDB, Dynamo DB, DocumentDB, Aurora Serverless, MySQL, SQLite, DuckDB
- **Other:** CUDA, Confluence, Rally, Azure DevOps, Jira, Git, GitHub, GitLab, Claude Code, Poolside.ai